Consolidated Teaching and Examination Scheme

B.Tech Chemical Engineering – 2024-28

MS: Mid Semester | IA: Internal Assessment | ES: End Semester | LW: Laboratory Work | LE/Viva: Laboratory Examination/Viva | CE: Continuous Evaluation | PP/NP: Passed/Not Passed

Semester 1													
Course Code	**Course Name**	**Theory**	**Tutorial**	**Practical**	**Hrs**	**Credits**	**Examination Scheme**					
							Theory			**Practical**		**Total**
							MS	**IA**	**ES**	**LW**	**LE/Viva**	
24MA101T	Mathematics - I	3	1	0	4	4	25	25	50	--	--	100
24HS101T	English Communication	2	0	0	2	2	25	25	50	--	--	100
24CP101T	Computer Programming - I	1	0	0	1	1	25	25	50	--	--	100
24CP101P	Computer Programming - I Laboratory	0	0	2	2	1	--	--	--	50	50	100
24CH101T	Engineering Chemistry	3	0	0	3	3	25	25	50	--	--	100
24CH101P	Engineering Chemistry Laboratory	0	0	2	2	1	--	--	--	50	50	100
24EE101T	Elements of Electrical and Electronics Engineering	3	0	0	3	3	25	25	50	--	--	100
24EE101P	Elements of Electrical and Electronics Engineering Laboratory	0	0	2	2	1	--	--	--	50	50	100
24BT101T	Biological Systems for Engineers	2	0	0	2	2	25	25	50	--	--	100
24HS104T	Organizational Behaviour	1	0	0	1	1	25	25	50	--	--	100
24YOG101 / 24NSS101 / 24NCC101	Yog, Health & Hygiene / National Service Scheme (NSS) / National Cadet Corps (NCC)	0	0	2	2	1	--	--	--	50	50	100
	Total	**15**	**1**	**8**	**24**	**20**						

Semester 2													
Course Code	**Course Name**	**Theory**	**Tutorial**	**Practical**	**Hrs**	**Credits**	**Examination Scheme**					
							Theory			**Practical**		**Total**
							MS	**IA**	**ES**	**LW**	**LE/Viva**	
24HS105T	Professional Communication	2	0	0	2	2	25	25	50	--	--	100
24MA103T	Mathematics - II	3	1	0	4	4	25	25	50	--	--	100
24CP102T	Computer Programming - II	1	0	0	1	1	25	25	50	--	--	100
24CP102P	Computer Programming - II Laboratory	0	0	2	2	1	--	--	--	50	50	100
24PH103T	Modern Physics	3	0	0	3	3	25	25	50	--	--	100
24PH103P	Modern Physics Laboratory	0	0	2	2	1	--	--	--	50	50	100
24ME101P	Workshop Practices	0	0	2	2	1	--	--	--	50	50	100
24ME102P	Engineering Graphics	0	0	4	4	2	--	--	--	50	50	100
24CV101T	Environmental Science	2	0	0	2	2	25	25	50	--	--	100
24HS102T	Universal Human Values	1	0	0	1	1	25	25	50	--	--	100
24HS103T	Indian Knowledge System	2	0	0	2	2	25	25	50	--	--	100
	Total	**14**	**1**	**10**	**25**	**20**						

24HS105T					Professional Communication					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
2	0	0	2	2	25	50	25	--	--	100

COURSE OBJECTIVES

- To strengthen the communication skills of professionals to make them ready for the modern workplace.
- To fine-tune their professional skills and expertise using communication skills.
- To participate in the lifelong learning process with confidence and certainty.
- To develop analytical, research and organizational skills through communication skills for a fulfilling career.

Unit I: TECHNICAL WRITING **Hours: 6**

E-mails; report writing; proposal writing (project, research and business); review writing (books and articles).

Unit II: ORATORY SKILLS **Hours: 6**

Art of introducing oneself; presentation skills; public speaking.

Unit III: EMPLOYABILITY SKILLS **Hours: 7**

Deciphering job descriptions; cover letter drafting; resume writing; group discussion; interview skills.

Unit IV: ADVANCED READING AND CRITICAL THINKING **Hours: 7**

Critical thinking; language and communication; ethical considerations; interdisciplinary perspectives; communication skills; philosophy of science.

Max 26 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Develop technical writing skills essential for effective communication in professional settings, including business environments, academic institutions and research organizations.
CO2 - Master the art of public speaking by delivering clear, engaging and persuasive presentations to diverse audiences.
CO3 - Demonstrate the ability to decipher job descriptions, extracting key requirements and qualifications to align their skills and experience effectively.
CO4 - Effectively prepare for and participate in group discussions, demonstrating active listening, critical thinking and coherent articulation of ideas.
CO5 - Understand the importance of clear, honest communication in engineering practice and ethical decision making.
CO6 - Explore interdisciplinary approaches to problem solving.

TEXT / REFERENCE BOOKS

1. Kumar, K. B. (2022). *Effective communication skills*. Khanna Book Publishing.
2. Wood, F. T. (2007). *Remedial English grammar*. Macmillan.
3. Sharma, S., & Mishra, B. (2009). *Communication skills for engineers and scientists*. PHI Learning.
4. Kaul, A. (2006). *Business communication*. Prentice-Hall of India.

24MA103T					Mathematics - II					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	1	0	4	4	25	50	25	--	--	100

COURSE OBJECTIVES

- To understand and apply principles of complex differentiation and integration effectively.
- To solve systems of equations, find eigenvalues and apply transformations confidently.
- To solve various ordinary differential equations, apply methods and tackle engineering problems.
- To analyze periodic functions, derive series and apply them in diverse applications.

Unit I: COMPLEX DIFFERENTIATION AND INTEGRATION **Hours: 16**

Limit, continuity and differentiability of a function of a complex variable; analytic functions; Cauchy-Riemann equations (in Cartesian coordinates); harmonic functions and their significance; singularities; definition of a complex line integral; contour integrals; Cauchy integral theorem; Cauchy integral formula (CIF); CIF for derivatives; Taylor and Laurent series; calculation of residues; Cauchy residue theorem; applications of residues to evaluate real definite integrals.

Unit II: MATRIX ALGEBRA AND ITS APPLICATIONS **Hours: 12**

Solution of a system of algebraic equations; rank of a matrix; consistency of a system of equations; characteristic equation of a square matrix; eigenvalues and eigenvectors of a real matrix; properties of eigenvalues and eigenvectors; Cayley-Hamilton theorem (without proof); finding the inverse of a matrix; diagonalization of a matrix.

Unit III: ORDINARY DIFFERENTIAL EQUATIONS WITH APPLICATIONS **Hours: 12**

Higher-order differential equations with constant coefficients; rules for finding C.F. and P.I.; method of variation of parameters; Cauchy and Legendre linear equations; linear differential equations of second order with variable coefficients; simultaneous linear equations with constant coefficients; applications of higher-order differential equations in solving engineering problems.

Unit IV: FOURIER SERIES **Hours: 12**

Periodic functions; odd and even functions; Euler's formulae for Fourier series in an interval of length 2π; change of interval; Dirichlet's conditions; half-range sine and cosine series; complex Fourier series; Parseval's identity and its applications.

Max 52 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Recall fundamental concepts of complex differentiation, matrix algebra, differential equations and Fourier series.
CO2 - Understand the significance of complex differentiation, integration, matrices, ODEs and Fourier series in respective contexts.
CO3 - Apply the concepts of complex functions, matrices, ODEs and Fourier series to extract solutions of engineering problems.
CO4 - Analyze the use of complex variables, matrices, ODEs and Fourier series in engineering problems.
CO5 - Assess the significance and effectiveness of mathematical concepts and theorems in solving real-world problems.
CO6 - Design and construct solutions to complex mathematical problems using transformations, mappings and advanced methods in differential equations.

TEXT / REFERENCE BOOKS

1. Churchill, R. V., & Brown, J. W. (2013). *Complex variables and applications* (9th ed.). McGraw-Hill.
2. Howie, J. M. (2003). *Complex analysis*. Springer-Verlag.
3. Jain, R. K., & Iyengar, S. R. K. (2016). *Advanced engineering mathematics* (5th ed.). Narosa Publishing House.
4. Kreyszig, E. (2015). *Advanced engineering mathematics* (10th ed.). John Wiley & Sons.
5. Strang, G. (2006). *Linear algebra and its applications* (4th ed.). Cengage Learning.
6. Hoffman, K., & Kunze, R. A. (2015). *Linear algebra* (2nd ed.). Prentice Hall of India.

24CP102T					Computer Programming - II					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
1	0	0	1	1	25	50	25	--	--	100

COURSE OBJECTIVES

- To impart the basic concepts of the Python programming language.
- To be familiar with data structures available in Python.
- To understand the concept of functions and object-oriented programming.
- To use file and exception handling for designing applications.

Unit I: INTRODUCTION AND LANGUAGE FUNDAMENTALS **Hours: 3**

Definition and historical background; comparison with other programming languages; Python IDEs; tokens - identifiers, keywords; data types - int, float, bool, complex, str, list, tuple, dictionary and set; type conversion; concepts of mutability, immutability and reusability; operators - arithmetic, relational, logical, identity and membership; input and output methods; control flow - conditional statements (if, elif, else), nesting; iterative statements (for, while, for-else, while-else); transfer statements (break, continue, pass).

Unit II: FUNCTIONS **Hours: 3**

Function declaration and calling; types of function arguments; scope of variables; recursive functions; types of functions - anonymous or lambda, map, filter, reduce; function aliasing; nested functions; decorator functions; decorator chaining; generator functions.

Unit III: OBJECT ORIENTED PROGRAMMING **Hours: 4**

Class, object, object reference; constructor and self variable; types of variables; types of methods; destructor; composition; aggregation; inheritance; polymorphism; abstract classes.

Unit IV: FILE AND EXCEPTION HANDLING **Hours: 3**

File handling - types of files, writing and reading with text files, important functions of file handling, working with directories, CSV file handling, pickling and unpickling; exception handling using try-except blocks; raising exceptions and custom exception classes.

Max 13 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand the basic concepts of programming with Python.
CO2 - Demonstrate proficiency in using different data types, operators and control structures in Python programs.
CO3 - Demonstrate the usage of both built-in and user-defined functions.
CO4 - Define and use classes in Python to represent attributes and methods, and demonstrate composition, aggregation, inheritance and polymorphism.
CO5 - Implement exception handling mechanisms to deal with errors.
CO6 - Demonstrate file operations and exception handling.

TEXT / REFERENCE BOOKS

1. Guttag, J. V. (2016). *Introduction to computation and programming using Python* (2nd ed.). Prentice Hall of India.
2. Downey, A., Elkner, J., & Meyers, C. (2002). *How to think like a computer scientist: Learning with Python*. Green Tea Press.
3. Sweigart, A. (2019). *Automate the boring stuff with Python* (2nd ed.). No Starch Press.
4. Brown, M. C. (2018). *Python: The complete reference*. McGraw-Hill.
5. Rao, R. N. (2018). *Core Python programming* (2nd ed.). Dreamtech Press.

24CP102P					Computer Programming - II Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

- To impart the basic concepts of the Python programming language.
- To be familiar with data structures available in Python.
- To understand the concept of functions and object-oriented programming.
- To use file and exception handling for designing applications.

LIST OF EXPERIMENTS

1. Language fundamentals: understanding interpretation, program execution flow, tokens, variables, keywords and identifiers, constants and variables.
2. Input-output operations: basic data types, mutability, immutability, operators, input/output statements, format specifiers, escape sequences.
3. Decision making and looping control structures: if, elif and else, nesting; iterative statements - for, while, for-else, while-else; transfer statements - break, continue, pass.
4. Advanced data types: string, list, tuples, dictionary, set.
5. Functions: basics of functions, types of arguments.
6. Types of functions: anonymous functions, function aliasing, nested functions, decorator functions and generator functions.
7. Object oriented programming: class, object, types of variables and methods, composition, aggregation, inheritance, types of inheritance.
8. File handling: writing and reading with text files and CSV files, working with directories, pickling and unpickling.

Max 26 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Apply the concepts of variables, decision making and looping with Python.
CO2 - Apply various data structures available in Python to solve computational problems.
CO3 - Implement advanced function concepts such as decorator chaining and generator functions.
CO4 - Design and implement object-oriented solutions using composition, aggregation, inheritance and polymorphism.
CO5 - Develop Python scripts for file handling tasks.
CO6 - Create and manage files in Python.

TEXT / REFERENCE BOOKS

1. Guttag, J. V. (2016). *Introduction to computation and programming using Python* (2nd ed.). Prentice Hall of India.
2. Downey, A., Elkner, J., & Meyers, C. (2002). *How to think like a computer scientist: Learning with Python*. Green Tea Press.
3. Sweigart, A. (2019). *Automate the boring stuff with Python* (2nd ed.). No Starch Press.
4. Rao, R. N. (2018). *Core Python programming* (2nd ed.). Dreamtech Press.

24PH103T					Modern Physics					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- To understand the concepts of modern physics.
- To explain the physics of electromagnetic waves and their propagation.
- To apply the acquired basic knowledge of solid state physics.
- To analyze the concepts of nuclear radiation physics and apply the fundamentals of modern physics to solve complex physical problems.

Unit I: MODERN PHYSICS **Hours: 8**

Review of quantum concepts: particle nature of light, photoelectric effect, Compton effect, matter waves, wave packets, phase and group velocity; interference, diffraction and polarization; engineering physics related applications; Davisson-Germer experiment; Heisenberg uncertainty principle.

Unit II: ELECTROMAGNETIC WAVES **Hours: 12**

Physical and mathematical concepts of gradient, divergence and curl; Gauss theorem and applications in gravitation and electrostatics; Stokes' theorem and engineering physics related applications; equation of continuity; Biot-Savart law; Ampere's law; Maxwell's equations; electromagnetic wave propagation; optical fibres and their applications in communication.

Unit III: SOLID STATE PHYSICS **Hours: 11**

Crystalline and amorphous solids; ionic crystals; covalent crystals; van der Waals bond; band theory of solids; semiconductor devices; electrical conductivity; resistivity; magnetism; superconductivity; introduction to BCS theory. Concepts of LASER: interaction of radiation with matter - quantum mechanical view, characteristics and types of lasers, engineering applications of lasers.

Unit IV: NUCLEAR RADIATION PHYSICS **Hours: 8**

Mass defect; binding energy; radioactivity; types of radiation; interaction of radiation with matter; radiation detectors; nuclear reactions; elements of nuclear reactors; fission and fusion; engineering physics related problems.

Max 39 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Relate the concepts of modern physics to shape the engineering perspective.
CO2 - Understand and analyse the physical phenomena underlying various physics concepts.
CO3 - Develop an analytical perspective towards physical problems.
CO4 - Understand the importance of the application of the topics studied.
CO5 - Explain and relate the importance of interdisciplinary problems.
CO6 - Strengthen a problem-solving attitude in physics using mathematical tools.

TEXT / REFERENCE BOOKS

1. Resnick, R., Halliday, D., & Krane, K. S. (2002). *Physics* (Parts I and II, 5th ed.). John Wiley & Sons.
2. Ghatak, A. (2017). *Optics* (6th ed.). Tata McGraw Hill.
3. Purcell, E. M. (2011). *Electricity and magnetism: Berkeley physics course* (Vol. 2, 2nd ed.). Tata McGraw-Hill.
4. Griffiths, D. J. (2015). *Introduction to electrodynamics* (4th ed.). Prentice Hall of India.
5. Avadhanulu, M. N., & Kshirsagar, P. G. (2018). *A textbook of engineering physics* (11th ed.). S. Chand & Company.
6. Feynman, R. P., Leighton, R. B., & Sands, M. (2012). *The Feynman lectures on physics* (Vol. 1). Narosa Publication.

24PH103P					Modern Physics Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

- To understand the characteristics of waves, semiconductors, heat pumps and LASERs.
- To enhance knowledge on the application of physics in engineering.
- To develop intellectual communication skills and discuss the basic understanding of various experimental principles involved.
- To demonstrate practical knowledge by applying experimental methods and analytical and graphical techniques to experimental data.

LIST OF EXPERIMENTS

1. To verify the inverse square law of radiation.
2. Study of photoconductivity.
3. Study of the Biot-Savart law and the magnetic field along the axis of a coil.
4. To determine the electrical conductivity of metals.
5. Study of the Hall effect.
6. Experiments on single-slit and double-slit diffraction and interference with a He-Ne laser.
7. Study of the I-V characteristics of a p-n diode.
8. Determination of thermal conductivity of different solids.
9. To measure the resistivity of a semiconductor by the four-probe method and determine the band gap.
10. Study of ferromagnetic hysteresis.
11. To study G.M. tube characteristics and to calculate the dead time.
12. Energy calibration of a CsI:Tl radiation detector and energy analysis of an unknown gamma source.
13. To determine the numerical aperture of a given fibre optic cable using far-field measurements.
14. Experiments with a heat pump.
15. Study of polarization of light using a LASER.

Max 26 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Analyse engineering problems and design components for their solution.
CO2 - Develop skills to utilize different tools for engineering problems.
CO3 - Analyse results and correlate them with theory and industrial applications.
CO4 - Design experimental set-ups and utilize them for component analysis.
CO5 - Identify problems and create solutions for research and development.
CO6 - Analyse scientific data and function efficiently as an individual and as a team member.

TEXT / REFERENCE BOOKS

1. Runyan, W. R. (1998). *Semiconductor measurements and instrumentation* (2nd ed.). McGraw Hill.
2. Sayer, M., & Mansingh, A. (2005). *Measurement, instrumentation and experiment design in physics and engineering*. Prentice Hall India.
3. Melissinos, A. C., & Napolitano, J. (2003). *Experiments in modern physics* (2nd ed.). Academic Press.
4. Nakra, B. C., & Chaudhry, K. K. (2016). *Instrumentation, measurements and analysis* (4th ed.). Tata McGraw Hill.
5. ORTEC. (2015). *Experiments in nuclear science: ORTEC laboratory manual*. ORTEC.

24ME101P					Workshop Practices					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

- To understand safety in various manufacturing processes.
- To learn how to use various measuring tools for engineering applications.
- To provide hands-on training and preparation of jobs using wood, metal and sheet as per drawing.
- To understand various manufacturing processes like machining, welding, soldering and 3D printing for prototypes.

LIST OF EXPERIMENTS

1. Introduction to workshop safety, layout and identification of various materials - plastic, wood, metals (ferrous and non-ferrous), rubber, glass, etc.
2. Use of measuring tools for engineering applications.
3. Fitting job: detailed drawing of the work piece, use of fitting tools and job preparation.
4. Carpentry job: detailed drawing of the work piece, use of carpentry tools and job preparation.
5. Sheet metal job: detailed drawing of the work piece, use of sheet metal working tools and job preparation.
6. Plumbing job: internal/external threading, piping network using tees, elbows, reducers and bends.
7. Hands-on training on a mini lathe and milling machine.
8. Demonstration of welding, brazing and soldering.
9. Soldering and desoldering for PCB.
10. 3D printing using polymer and metal.

Max 26 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Define workshop safety and various engineering materials.
CO2 - Understand various measuring equipment.
CO3 - Apply various workshop tools in preparing jobs for carpentry, fitting, sheet metal and plumbing.
CO4 - Examine various manufacturing operations like welding and machining.
CO5 - Evaluate soldering operations for PCB.
CO6 - Create a prototype using 3D printing.

TEXT / REFERENCE BOOKS

1. Hajra Choudhury, S. K., & Hajra Choudhury, A. K. (2010). *Elements of workshop technology* (Vols. 1-2). Media Promoters and Publishers.
2. Bawa, H. S. (2009). *Workshop practice*. Tata McGraw-Hill.
3. Kalpakjian, S., & Schmid, S. R. (2014). *Manufacturing engineering and technology* (7th ed.). Pearson Education India.
4. Rao, P. N. (2018). *Manufacturing technology* (Vols. 1-2, 4th ed.). Tata McGraw Hill.

24ME102P					Engineering Graphics					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	4	2	4	--	--	--	50	50	100

COURSE OBJECTIVES

- To cover the fundamentals of engineering drawing and standards used in drawing.
- To enable students to communicate ideas using orthographic and isometric projection methods.
- To help students use CAD software to prepare drawings.
- To demonstrate the presentation of drawings using sketching and 3D modelling in a CAD tool.

LIST OF EXPERIMENTS

1. Introduction to engineering graphics: principles of engineering graphics and their significance, drawing instruments and accessories, lettering and numbering, types of lines, dimensioning methods, basic geometric drawing, reading a drawing.
2. Orthographic projection: introduction to projection, types of projection, first angle and third angle projection.
3. Isometric projection: principles of isometric projection, isometric scale, isometric views, conventions, conversion of isometric views to orthographic views and vice versa.
4. Projection of solids and development of surfaces: classification of solids, projections of solids such as cylinder, cone, pyramid and prism with inclination to reference planes, development of surfaces of right regular solids.
5. Introduction to computer aided engineering drawing: theory of CAD software, user interface, commands, coordinate system, menus and toolbars, planes, dimensioning, saving of files, select and erase objects, zoom tools.
6. Basic sketching using a CAD tool: sketch entities using origin, points, lines, circles, arcs, polygons, fillets and chamfers, trim, extend and offset, copy, cut and delete.
7. Advanced sketching using a CAD tool: sketching entities using relation constraints, mirror, patterning, full definition of drawing.
8. Basic 3D modelling using a CAD tool: extrude, cut, drawing on different planes, editing, symmetric and revolving features.
9. Computer aided drawing sheets: preparing drawing sheets, creating different views, section views and drawing templates.

Max 52 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Recall the fundamentals of engineering graphics by considering basic rules of drawing, dimensioning and labelling.
CO2 - Explain the principle of projection using orthographic and isometric projection.
CO3 - Represent 2-dimensional drawings using a CAD tool.
CO4 - Construct 3-dimensional geometries using a CAD tool.
CO5 - Apply the concepts of engineering drawing by organizing drawing views and applying necessary dimensions while preparing drawing sheets.
CO6 - Analyse the intricate details of solids using projection of solids, sectioning of solids and development of lateral surfaces.

TEXT / REFERENCE BOOKS

1. Bhatt, N. D., Panchal, V. M., & Ingle, P. R. (2014). *Engineering drawing* (53rd ed.). Charotar Publishing House.
2. Shah, P. J. (2014). *Engineering graphics*. S. Chand Publishing.
3. Agrawal, B., & Agrawal, C. M. (2013). *Engineering drawing* (2nd ed.). Tata McGraw Hill.
4. Hanifan, R. (2014). *Perfecting engineering and technical drawing.* Springer International Publishing.

24CV101T					Environmental Science					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
2	0	0	2	2	25	50	25	--	--	100

COURSE OBJECTIVES

- To develop a comprehensive perspective of the environment and sustainable development.
- To understand the causes and effects of various types of pollution.
- To develop an understanding of various strategies for controlling pollution.
- To introduce the emerging environmental domains.

Unit I: INTRODUCTION TO ENVIRONMENT **Hours: 5**

Sustainable development; sustainable development goals; environmental studies - importance and multidisciplinary nature; introduction to environmental parameters and their standards (air, water, soil, noise); ecosystem and its types; ideal ecosystem; biodiversity - its importance and conservation.

Unit II: MULTI-SCALE ENVIRONMENTAL POLLUTION (GLOBAL, REGIONAL AND LOCAL) **Hours: 6**

Pollution: causes and effects of different types of pollution - air pollution, water pollution, soil pollution, solid waste (organic and inorganic) pollution, hazardous waste pollution, marine pollution, noise pollution, thermal pollution, radioactive pollution; introduction to man-made disasters such as floods, heat waves and landslides; introduction to various instruments for measuring air pollution, water pollution and noise.

Unit III: ENVIRONMENTAL POLLUTION CONTROL STRATEGIES **Hours: 8**

Multi-approaches for reducing various types of pollution; introduction to water and wastewater treatment technologies; air and noise pollution control techniques; introduction to different environmental management concepts such as Swachh Bharat Mission and Mission LiFE; Indian culture and traditional wisdom for managing the environment.

Unit IV: EMERGING ENVIRONMENTAL MANAGEMENT DOMAINS **Hours: 7**

Concept of Zero Liquid Discharge (ZLD) and reuse of treated wastewater; Green Credit Rules 2023; Clean Development Mechanisms (CDM) and carbon credits; green buildings; carbon footprint and water footprint; green business; international environmental laws; environmental auditing.

Max 26 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Demonstrate comprehension of sustainable development and environmental aspects.
CO2 - Recognize the interdisciplinary characteristics inherent in environmental studies.
CO3 - Evaluate the impact of various pollutants on the environment.
CO4 - Assess the efficacy of different technologies for environmental pollution control.
CO5 - Analyze different environmental management policies and their implications.
CO6 - Synthesize knowledge about emerging environmental management paradigms.

TEXT / REFERENCE BOOKS

1. Bharucha, E. (2013). *Textbook for environmental studies*. University Grants Commission.
2. Botkin, D. B., & Keller, E. A. (2011). *Environmental science: Earth as a living planet* (8th ed.). Wiley India.
3. Miller, T. G. (2006). *Environmental science* (11th ed.). Cengage Learning.
4. Rajagopalan, R. (2015). *Environmental studies: From crisis to cure* (3rd ed.). Oxford University Press.
5. Masters, G. M., & Ela, W. P. (2013). *Introduction to environmental engineering and science* (3rd ed.). PHI Learning.

24HS102T					Universal Human Values					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
1	0	0	1	1	25	50	25	--	--	100

COURSE OBJECTIVES

- To understand the need for nurturing human values through the process in a value-based education system.
- To understand and develop a holistic perspective on self-exploration and being in harmony with family, society and nature.
- To facilitate students in understanding harmony at all levels and applying it in their profession and workplace to lead an ethical life.
- To appreciate the scope of eco-friendly systems and a universal human order.

Unit I: HUMAN VALUES AND PROCESS OF VALUE EDUCATION **Hours: 3**

Human values, human aspirations and the ultimate goal; understanding happiness and prosperity; appraising the meaning of satisfaction and happiness in the current scenario; harmony and compatibility; values imbibed in the education system and process.

Unit II: KNOWING SELF - HARMONY WITH SELF **Hours: 4**

Understanding self, capabilities and challenges; understanding material (physical facilities) and spiritual needs - needs of mind and body; understanding the body as an instrument; harmony between mind and body; synchronizing physical and mental health; practising healthy habits.

Unit III: HARMONY IN RELATIONSHIP - FAMILY, SOCIETY AND NATURE **Hours: 3**

Harmony in relationships; values for harmony in human-human interaction; harmony in family and society; trust and respect for others; self-esteem and ego; equality, equity, inclusion and liberation; concept of Vasudhaiva Kutumbakam; understanding co-existence and sync with nature.

Unit IV: HARMONY IN PROFESSION AND ETHICAL BEHAVIOUR **Hours: 3**

Ethical human conduct; acceptance and respect; appraising the qualities of others; professional competence for enabling harmony in systems and enabling a universal human order; scope of eco-friendly systems; strategies to reach a harmonious ecosystem and Sarvabhauma Vyavastha.

Max 13 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand the significance of human values, their need and the process of value education.
CO2 - Appraise the meaning of happiness and prosperity as short- and long-term goals of life in the context of the current scenario.
CO3 - Distinguish between the mind and body, and physical and spiritual well-being, for harmony within the self.
CO4 - Assess the value of harmonious relationships based on trust and respect and their role in building a harmonious society.
CO5 - Understand the importance of harmony with nature and appreciate co-existence for a harmonious ecosystem.
CO6 - Create an ethical professional workplace and strategize to uphold human values at all levels and interactions.

TEXT / REFERENCE BOOKS

1. Gaur, R. R., Sangal, R., & Bagaria, G. P. (2019). *A foundation course in human values and professional ethics* (2nd ed.). Excel Books.
2. Nagraj, A. (2003). *Jeevan Vidya: Ek parichay*. Divya Path Sansthan.
3. Tripathi, A. N. (2009). *Human values* (2nd ed.). New Age International Publishers.
4. Gandhi, M. K. (2018). *The story of my experiments with truth*. Fingerprint Publishing.
5. Schumacher, E. F. (2011). *Small is beautiful: A study of economics as if people mattered*. Blond & Briggs.

24HS103T					Indian Knowledge System					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
2	0	0	2	2	25	50	25	--	--	100

COURSE OBJECTIVES

- To connect with the tradition of the Indian Knowledge System (IKS).
- To enable students to define the philosophical foundations of IKS.
- To understand the fundamentals of art and architecture, traditional and historical town planning and ancient Indian art and architecture.
- To relate IKS with engineering, mathematics and other disciplines.

Unit I: INTRODUCTION TO IKS **Hours: 6**

What is IKS? Historical and philosophical foundations; knowledge framework and classifications in IKS; Indian scheme of knowledge; the knowledge triangle; framework for establishing valid knowledge; deductive and inductive logic frameworks; potential fallacies in the reasoning process.

Unit II: IKS AND ENGINEERING DISCIPLINES **Hours: 6**

Engineering marvels of ancient India; civil engineering - urban planning and water management; materials engineering - sustainable materials and traditional construction techniques, material science and metallurgy in IKS; mechanical engineering - robotics and automation in ancient India; the role of IKS in addressing the climate change challenge; IKS principles for resource management and conservation.

Unit III: IKS AND MATHEMATICS **Hours: 7**

IKS and Indian mathematics; algebra; geometry; trigonometry; binary mathematics; magic squares; the Indian calendar system; astronomical instruments (yantras).

Unit IV: IKS AND OTHER DISCIPLINES **Hours: 7**

Indian administration; concept of state; Anushasan Parva of the Mahabharata; social and political philosophies of ancient India; IKS and medicine - Ayurveda, a holistic approach to health and well-being.

Max 26 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Define the fundamental concepts of the Indian Knowledge System.
CO2 - Describe various concepts of IKS relevant from an engineering perspective.
CO3 - Apply the Indian mathematical framework, including numbers, spherical trigonometry and the celestial sphere.
CO4 - Analyse the science of astronomy and the different units of time discussed in the texts.
CO5 - Weigh the IKS systems vis-à-vis current applications in various spheres of understanding.
CO6 - Design applications and processes by incorporating traditional knowledge to address present-day concerns.

TEXT / REFERENCE BOOKS

1. Mahadevan, B., Pavana, N., & Bhat, V. R. (2022). *Introduction to Indian knowledge system: Concepts and applications*. PHI Learning.
2. Saraswati, S. C. (2008). *The Vedas*. Bharatiya Vidya Bhavan.
3. Kapoor, K., & Singh, A. K. (2005). *Indian knowledge systems* (Vols. 1-2). Indian Institute of Advanced Study.